



Internet of Things Part I – Final Project

Smart Watering

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Introduction

In the context of **Precision Agriculture**, we propose a shift in field management:

1. From Static to Dynamic

- Shift from fixed timers (e.g., 8:00 AM) to **Real-Time Monitoring**.
- Dynamic control driven by environmental data.

2. Automated Decision Making

- Autonomous activation based on **critical thresholds**.
 - **Goal:** Maximize crop health, minimize water waste.
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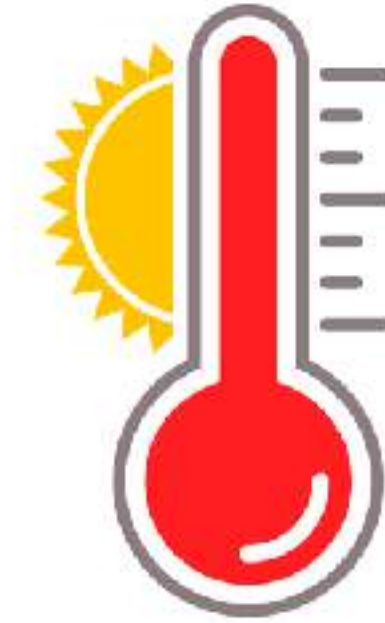


Goal

Define a sensor network capable of controlling the activation and deactivation of field sprinklers based on two **environmental parameters**:

- **Heat (C):** Measured using temperature in Degrees Celsius ($^{\circ}\text{C}$).
- **Precipitation (P):** Measured in Millimeters per hour (mm/h).

Control Logic: If Temperature exceeds a threshold and Rainfall is insufficient to compensate, the sprinklers will be activated.



Simulation Strategy

Dual-Event Modeling: Two analog events are used to represent physical variables:

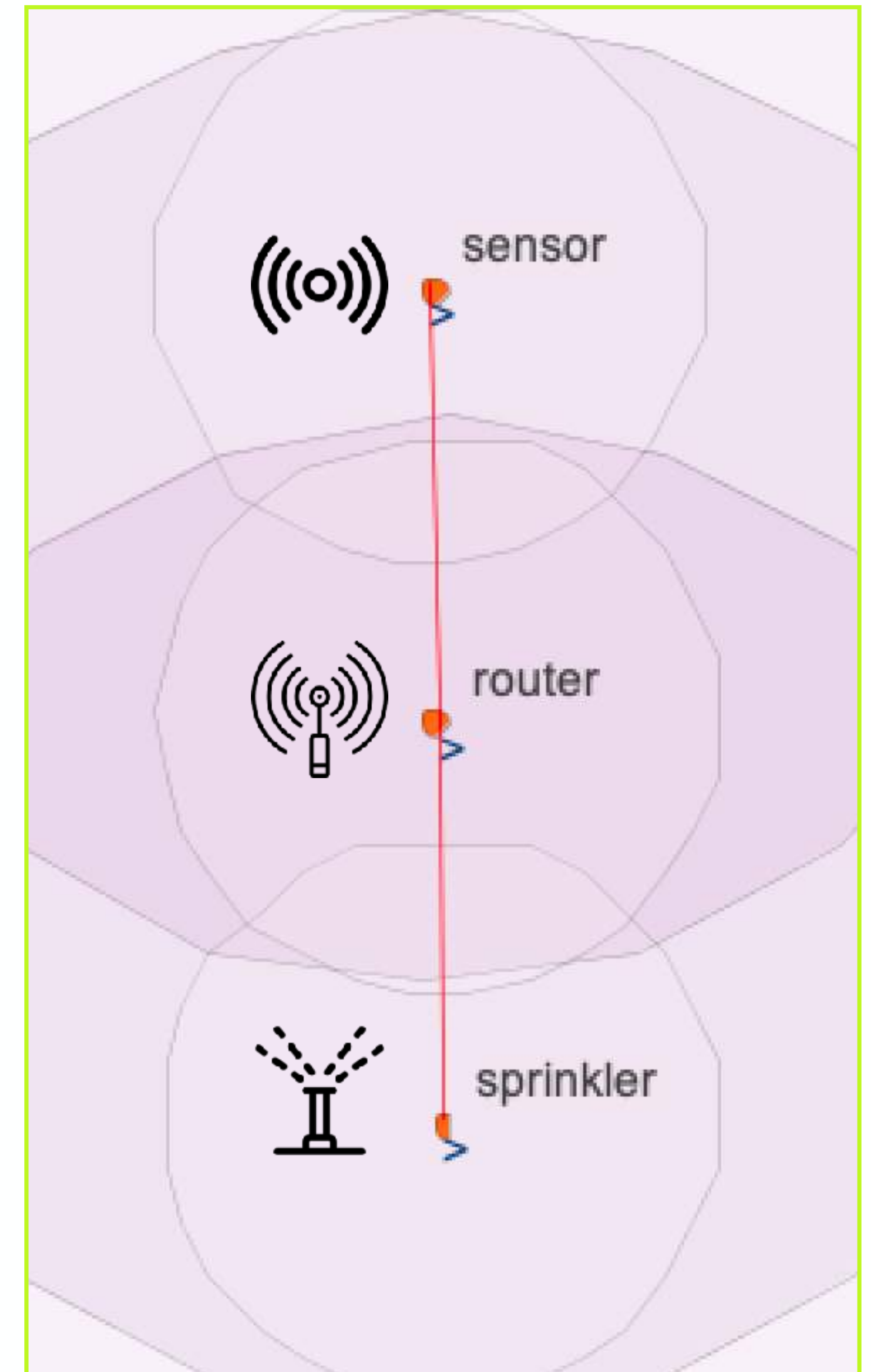
- **Gas ID 0:** Simulates Temperature field (C).
- **Gas ID 1:** Simulates Rainfall intensity (P).
- **Activation Thresholds:** A sensor determines whether irrigation should be activated and issues a trigger message IF and ONLY IF:

$$C > 28\text{ }^{\circ}\text{C AND } P < 5\text{ mm/h}$$

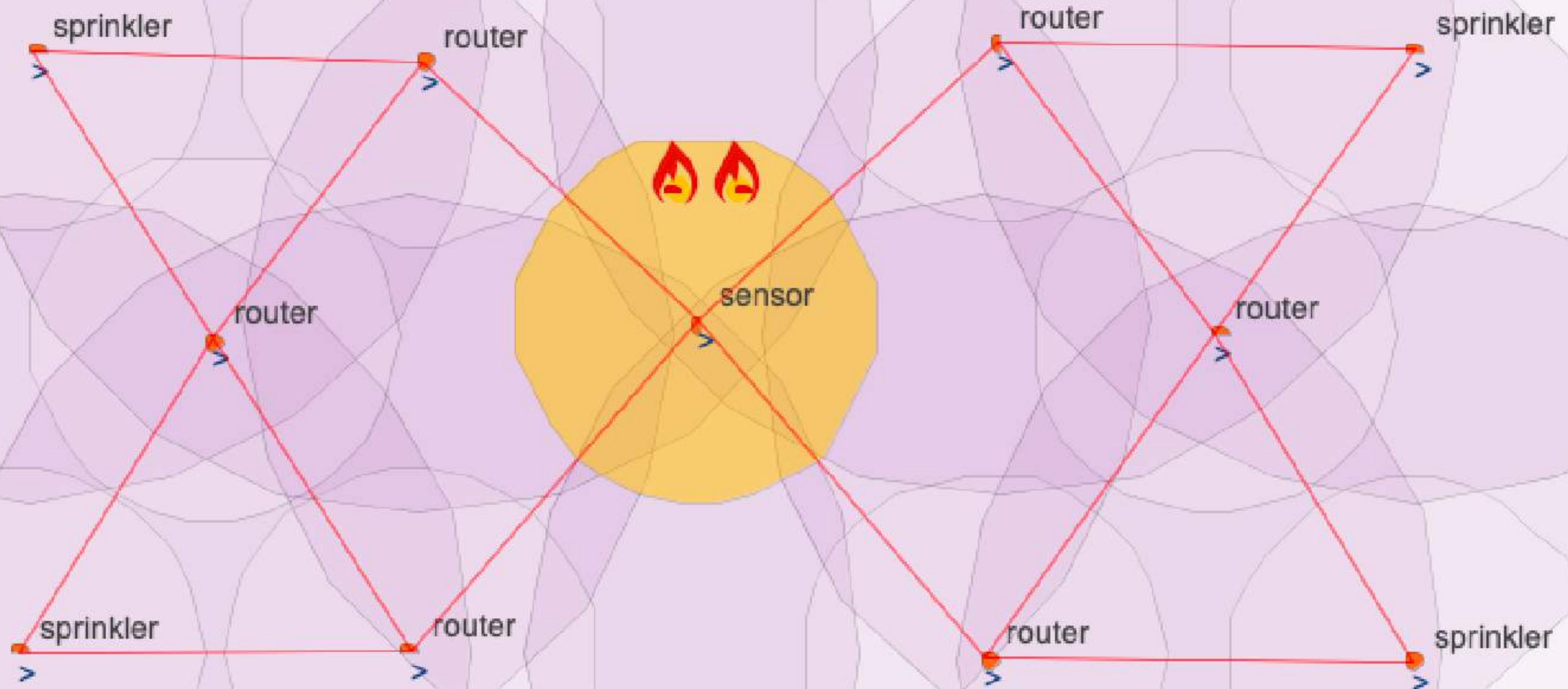


Node Roles

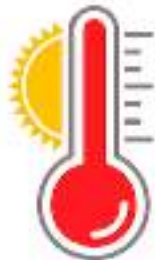
- **Sensor:** Detects Temperature and Rainfall. It holds the decision logic.
- **Router:** Acts as a relay, forwarding ON/OFF commands to remote nodes.
- **Sprinkler:** The final receiver. It **starts** or **stops** irrigation according to the received instructions.



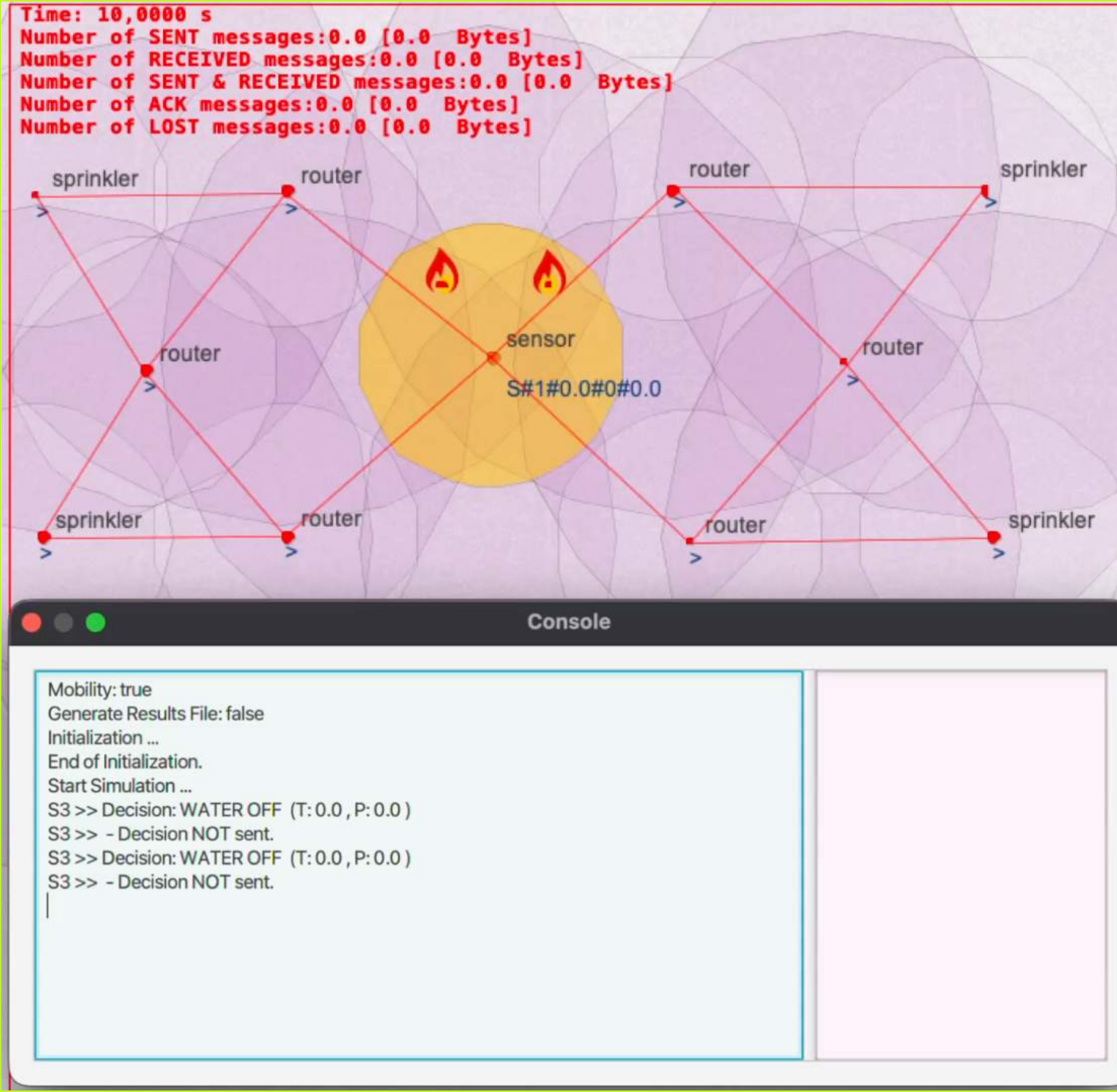
Network on cupCarbon



Demo

		Sensor Decision
28.5	1.0	ON
27.5	1.0	OFF
26.0	4.0	/
30.0	4.0	ON
30.5	6.0	OFF

N.B. Decision Rule: $C > 28$ AND $P < 5$



Natural Events

heat.gas

0	15.5
10	28.5 
10	27.5
10	26.0
10	30.0 
10	30.5 
10	32.0 
10	27.5
10	29.0 
10	22.5
10	25.0
10	29.0 
10	30.0 
10	27.0
10	33.0 
10	25.0
10	25.0

rain.gas

0	0.0
10	1.0
10	1.0
10	4.0
10	4.0
10	6.0 
10	5.5 
10	7.0 
10	0.0
10	0.0
10	1.0
10	0.0
10	1.0
10	3.0
10	6.0 
10	5.0 
10	0.0

sensor.csc

```
# --- SENSOR LOGIC ---

# Activation Thresholds Variables
# T > 28°C triggers irrigation, P < 5mm/h allows irrigation
set th_T 28
set th_P 5

# State Variable: Tracks the last sent command (0 or 1)
# Prevents sending redundant messages to save battery
set lastDecision 0

loop
  # Sensing
  areadsensor raw
  print raw

  # Initialize local variables to a control value (99)
  # This helps detect if a specific sensor reading failed
  set my_temp 99
  set my_rain 99

  if(raw != "X")
    rdata raw type id1 v1 id2 v2
    # Mapping GAS ID
    if(id1 == 0)
      set my_temp v1
    end
    if(id1 == 1)
      set my_rain v1
    end
    if(id2 == 0)
      set my_temp v2
    end
    if(id2 == 1)
      set my_rain v2
    end
  end
end
```

```
# Check validity: Proceed only if both sensors responded (value != 99)
if((my_temp != 99) && (my_rain != 99))

  # LOGIC: High Temperature AND Low Rain required to Water
  if((my_temp > th_T) && (my_rain < th_P))
    data msg 1
    cprint "Decision: WATER ON (T:" my_temp ", P:" my_rain ")"
  else
    data msg 0
    cprint "Decision: WATER OFF (T:" my_temp ", P:" my_rain ")"
  end

  # Send command ONLY if the state has changed
  if (lastDecision != msg)
    set lastDecision msg
    send msg
    cprint " - Decision sent"
  else
    cprint " - Decision NOT sent."
  end
else
  cprint "T or P are missing (T:" my_temp ", P:" my_rain ")"
end

end

delay 5000
```


router.csc

```
# --- ROUTER NODE ---

# Memory Variable: Stores the last relayed packet
# Initialized to a dummy value (3)
set last_msg 3

loop
    receive packet

    # Check: Is this a new message?
    # If not, it's an echo -> DROP IT
    if (packet != last_msg)
        # Forwarding and Updating mamory
        send packet
        set last_msg packet
    end

    delay 200
```


sprinkler.csc

```
# --- SPRINKLER NODE ---

# variables for console printing
atget id myid
set last_cmd 3
loop
    # Check messages and extract the Command Value
    receive msg
    rdata msg cmd

    # ON/OFF Control Logic
    if(cmd != "X")
        if(cmd != last_cmd)
            set last_cmd cmd
            # CASE: OPEN VALVE (Water ON)
            if(cmd == 1)
                mark 1
                cprint ">>>Sprinkler " myid ": IRRIGATION ON <<<"
            end

            # CASE: CLOSE VALVE (Water OFF)
            if(cmd == 0)
                mark 0
                cprint ">>>Sprinkler " myid ": IRRIGATION OFF <<<"
            end
        end
    else
        # Error handling for empty or corrupted packets
        cprint "ERROR: Empty Variable!"
    end
    delay 200
end
```

Energy Efficiency Analysis

Long-Term Sustainability

- Estimated Lifespan: **> 5 Years**
- Degradation Factor: Includes **3% Annual Self-Discharge** (realistic chemical aging).

Operational Parameters (Simulation Setup)

- **Duty Cycle:** Sensing every **1 minute**.
 - **Transmission Logic: Event-Based.**
 - Data is not sent continuously but triggered by meteorological events.
 - *Result:* Significant reduction in radio usage compared to continuous streaming.
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Energy Efficiency Analysis

Sensitivity Analysis (Scalability)

The lifetime results are dynamic and depend on specific deployment needs:

- **Field Size (Coverage):** Smaller fields allow for **Adaptive Power Control (atpl)**, reducing Etx and extending life further.
 - **Temporal Resolution:** Increasing sampling frequency (e.g., **30s interval**) would linearly increase consumption, reducing lifetime to 3-4 years.
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Network Evaluation

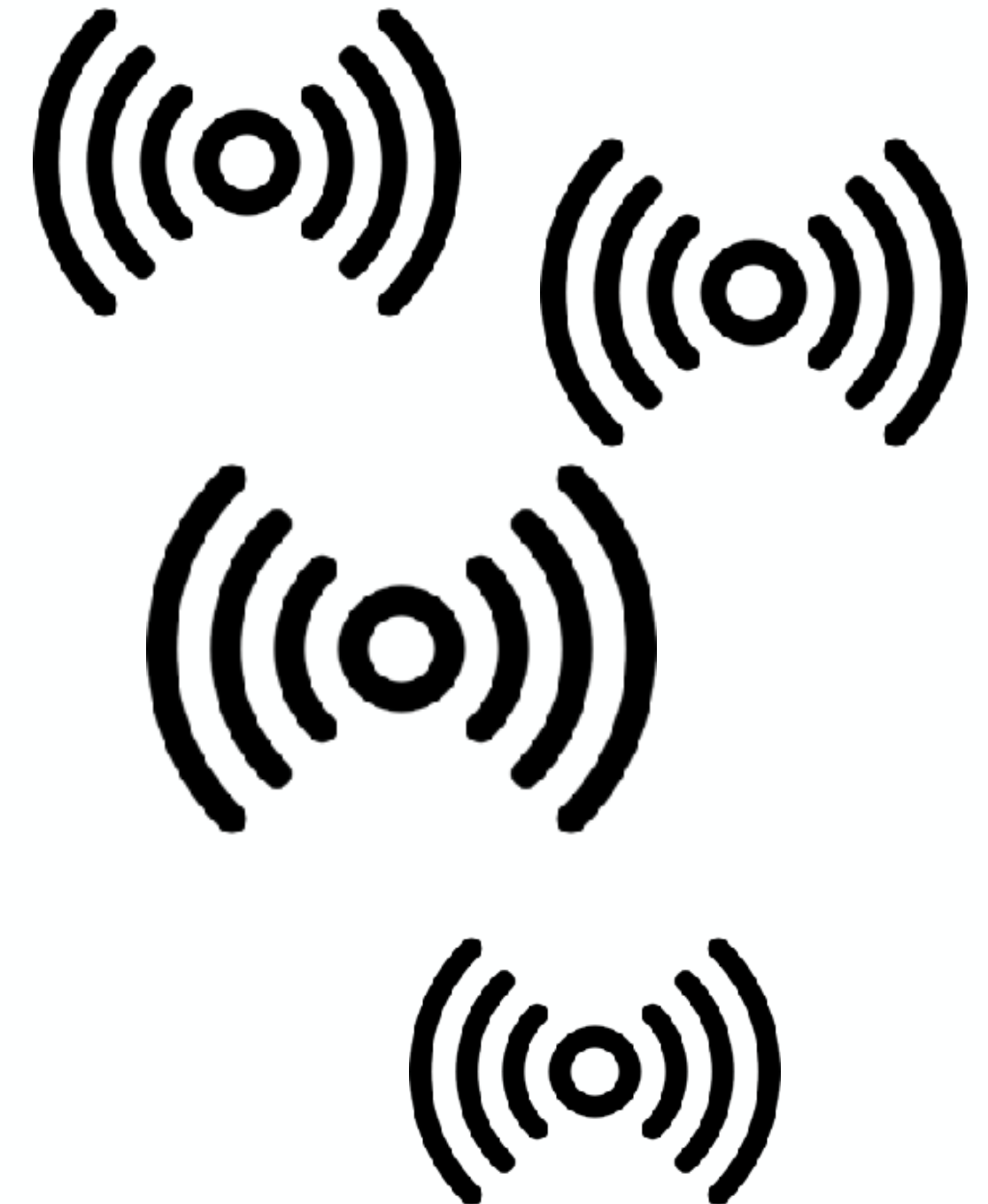
Edge Logic Strategy

- **Edge Computing:** Decision-making logic is assigned to the sensor node rather than the sprinklers.
 - Event-Driven Transmission:
 - Sensors transmit data **only upon state change** (e.g., from OFF to ON).
 - **Redundancy Elimination:** Prevents the transmission of repetitive, duplicate messages (e.g., continuous "ON-ON-ON" streams).
 - **Outcome:** Drastic reduction in radio traffic and energy consumption, significantly extending **Network Lifetime**.
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Risks Evaluation

Sensor Risk Analysis:

- **Single Point of Failure:** Relying on a single sensor centralizes the risk.
- **Failure Scenario:** If the **sensor** malfunctions or the battery drains, the entire actuation logic fails due to a lack of hardware redundancy.



Risks Evaluation

Scalability & Use Cases: Network topology must adapt to the physical scale of the deployment:

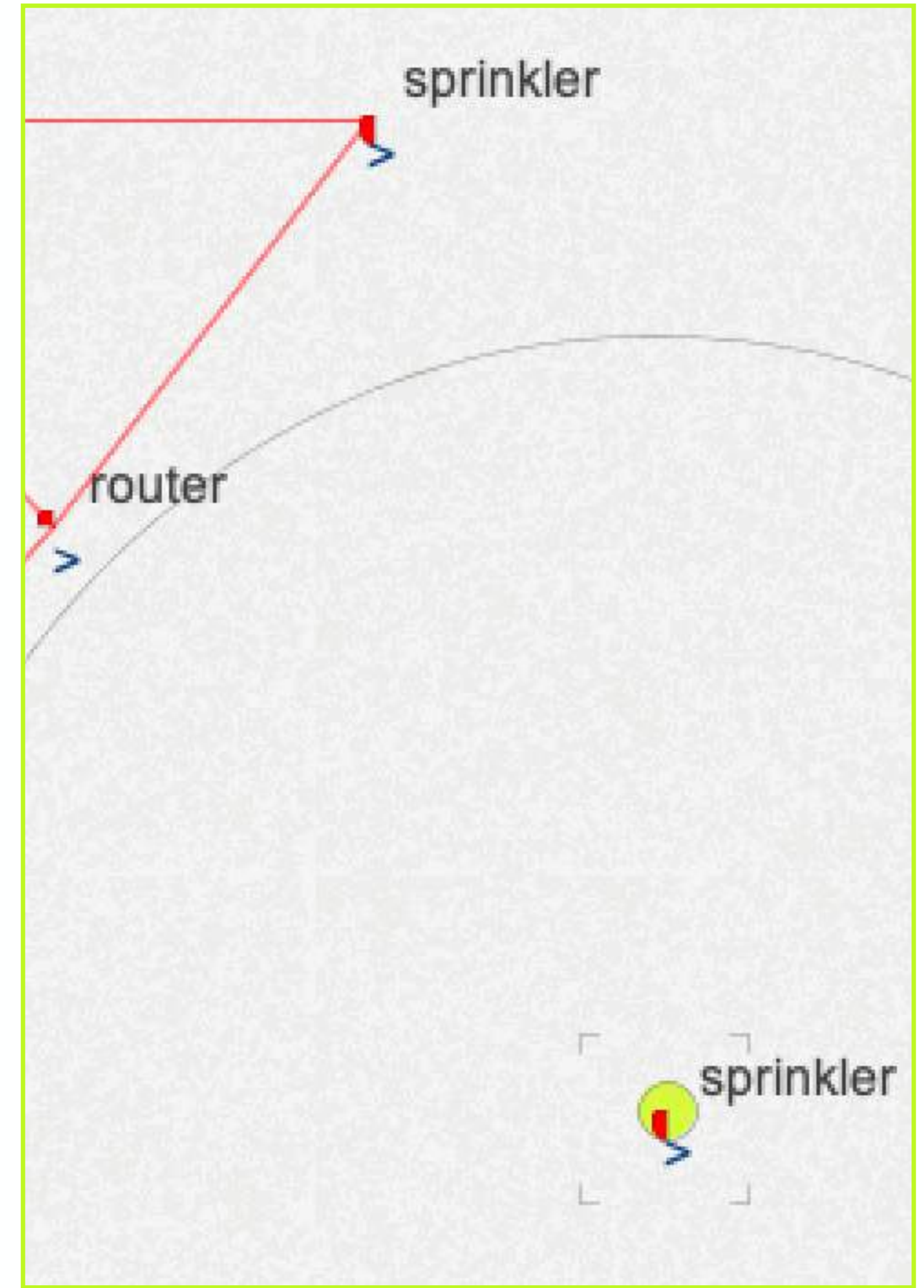
- **Small Scale Scenario** (e.g., Home Garden):
A single sensor is acceptable (Low cost, easy maintenance, protected environment).
- **Large Scale Scenario** (e.g., Agricultural Fields):
Redundancy (multiple sensors) is mandatory to ensure reliability and coverage over vast areas.



Risks Evaluation

Sprinkler Risk Analysis:

- **Scenario:** A sprinkler node receives an ON command but subsequently loses connection (e.g., interference, router failure, battery death).
- **Consequence:** The "OFF" command is never received. The sprinkler remains stuck in the open state, causing flooding, crop damage, and water resource depletion.



Risks Evaluation

The Solution: Local Failsafe Logic

- **A possible implementation** can be a local timer on the actuator node.
- **Keep-Alive Protocol:** The central logic should send a "Refresh" command periodically to keep the valve open.
- **Auto-Shutdown:** If the sprinkler does not receive a valid message within a specific safety window, it **automatically switches to OFF**, prioritizing safety over operation.

```
// -- PSEUDOCODE SNIPPET --  
  
set safety_timer 0  
loop  
  read msg  
  if(msg == "ON")  
    set safety_timer 0  
    mark 1                //Sprinkler ON  
  end  
  
  inc safety_timer        //Increment  
  if(safety_timer > 60)   //Th: 60 cycles  
    mark 0                //FORCE OFF  
  end  
  delay 1000
```

Summary & Key Takeaways

Successfully shifted from static scheduling to **Dynamic Precision Watering**.



Intelligent Edge Control

Processing decisions locally on the sensor node to **minimize** latency and **data traffic**.



Safe Architecture

Reliable Routing: Implemented **Controlled Flooding** to ensure coverage without broadcast storms.



Energy Sustainability

Through **Event-Driven Transmission** and aggressive Duty Cycling, I achieved a realistic network lifetime of **> 5 Years**.



Scalable Deployment

Flexible Topology: The architecture supports scalability from single-node domestic gardens to multi-hop Mesh networks for large agricultural fields



Thanks for the attention
